Summary in Graph

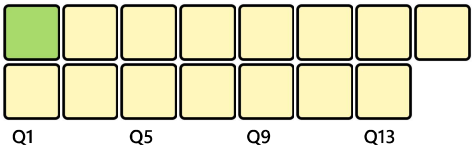
Exam Summary (GO Classes DA Test Series 2026 | Machine Learning | Topic Wise Test 9)

Qs. Attempted:	1 (1 + 0)	Correct Marks:	1 (1 + 0)
Correct Attempts:	1 (1 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	1 (1 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	45 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



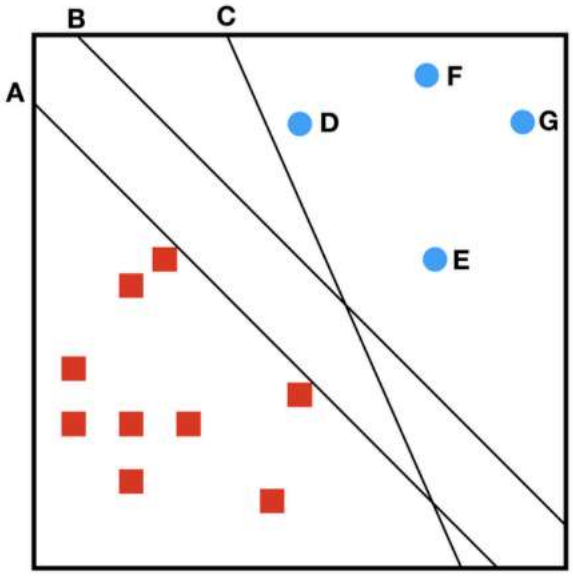
* Average Stats based on first-attempt stats of Top 10% (17) out of 171 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Which (if any) of the decision boundaries could be returned by perceptron (with a bias parameter) after training until convergence on this dataset? Mark all that apply.



A. A
B. B
C. C
D. None of the above

Your Answer: A;B;C | Correct Answer: A;B;C | Correct | Time taken: 38min 27sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Suppose a random variable has the following probability mass function:

$$P(X = k) = (k-1)p^2(1-p)^{k-2} \text{ for } k = 2, 3, 4, \ldots$$

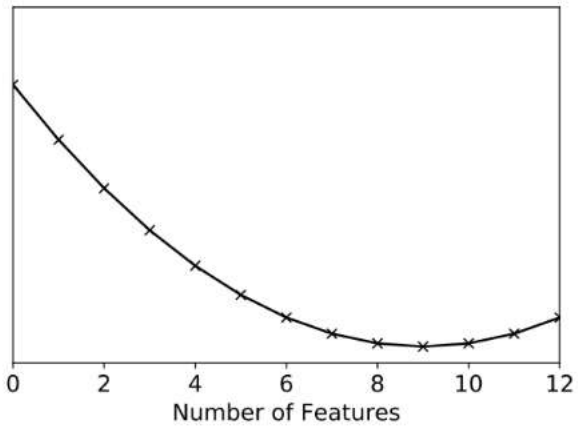
We observe 3 independent values from the pmf to be 3,7,2 What is the likelihood for p ?

A. $12p^2(1-p)^6$
B. $12p^6(1-p)^6$
C. $54p^6(1-p)^{12}$
D. $12p^8(1-p)^5$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 07min 29sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

In the process of training linear models with different numbers of features, you created a plot but forgot to include the Y -axis label. Which of the following(s) could the Y -axis represent? Select all that apply.



A. Training error
B. Bias
C. Test error
D. Variance

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider the following design matrix:

a	b	c
1	0	0
1	0	0
0	0	1
0	1	0

Suppose we train a linear model M_1 on this data. Then, we replace all the 1 values in column a with 3's and all the 1 values in column b with 2's and train a new linear model M_2. Neither M_1 nor M_2 have a bias or intercept term.
On the training data, the average squared loss for M_1 will be:

A. greater than that of M_2.
B. less than that of M_2.
C. equal to that of M_2.
D. impossible to tell.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Suppose we have a design matrix X with n observations, d features, and an intercept term. We tune and evaluate a regularized linear regression model using two hyperparameters, λ_1 and λ_2. There are i choices for λ_1 and j choices for λ_2, leading to $i \cdot j$ combinations of hyperparameters. For a single combination of hyperparameter values, how many model parameters do we fit? Assume k fold cross validation is used.

A. $d + 1$
B. $k \cdot (d + 1)$
C. $i \cdot j \cdot (d + 1)$

D. $\frac{n}{k}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider the following binary classification dataset consisting of four training examples, each with two features, x1 and x2:
1. $(-1, -1, +1)$
2. $(-1, -1, +1)$
3. $(+1, +1, -1)$
4. $(+1, +1, -1)$
Consider the following two statements:

Statement A: The perceptron algorithm with no bias term will converge on this dataset.

Statement B: This dataset can be perfectly classified using a linear classifier with no bias term.

Which of the following is correct?

A. Both A and B are true.
B. Only A is true.
C. Only B is true.
D. Both A and B are false.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Suppose that $X_1, \ldots, X_m$ are binary categorical input attributes and Y is a categorical output attribute. We plan to learn a decision tree without pruning, using the standard algorithm. Assume attributes are binary.

Statements:

1. The maximum depth of the decision tree must always be less than $m + 1$.
2. For data with R sample points, the maximum depth of the decision tree must always be less than $1 + \log_2(R)$.

Which of the following is correct?

A. Only Statement 1 is true.
B. Only Statement 2 is true.
C. Both Statement 1 and Statement 2 are true.
D. Neither Statement 1 nor Statement 2 is true.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

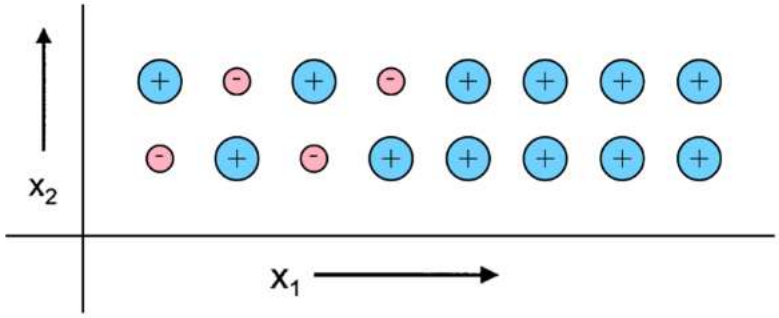
Consider an SVM classifier using two different feature transformations:
- The first feature transformation maps each input data x to $\Phi_1(x) = [x_1, x_2]$.
- The second feature transformation maps each input data x to $\Phi_2(x) = [2x_1, 2x_2]$.
How does the margin attained using $\Phi_2(x)$ compare to the margin attained using $\Phi_1(x)$?

A. Greater
B. Equal
C. Smaller
D. Any of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

The figure below shows a dataset with two inputs, X_1 and X_2, and one output, Y , which can be either positive (+) or negative (-). The dataset has 16 data points: 12 are labeled as positive (+) and 4 are labeled as negative (-).

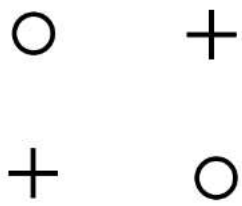


If the algorithm OVERFIT creates a decision tree without any pruning, how many leaf nodes will the decision tree have for this dataset?

Your Answer: | Correct Answer: 9 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Consider the following data set:



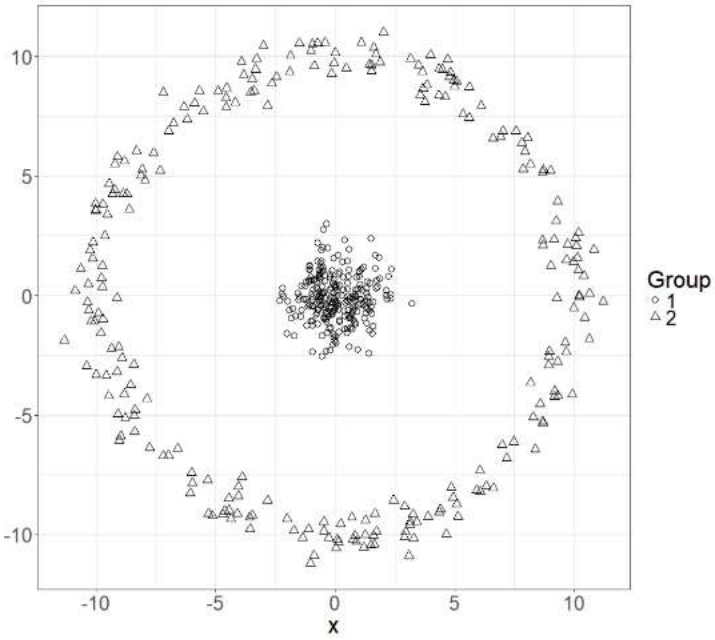
Circle all of the classifiers that will achieve zero training error on this data set.

A. Logistic regression
B. SVM (linear kernel)
C. Depth-2 decision trees
D. 3-NN classifier

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider the data plotted below. Which of the following clustering methods is most likely to return the groupings below using untransformed x and y values? Select only one.



A. Single-linkage clustering
B. Complete-linkage clustering
C. Average-linkage clustering
D. k-means clustering

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider ridge regression models defined as follows:

$$\hat{\theta_A} = \arg\min_{\theta} \frac{1}{n} \sum_{i=1}^{n} \left(y_i - \mathbb{X}_i^T \theta\right)^2 + \lambda \sum_{k=0}^{p} \theta_k^2$$

$$\hat{\theta_B} = \arg\min_{\theta} \frac{1}{n} \sum_{i=1}^{n} \left(y_i - \mathbb{X}_i^T \theta\right)^2 + \lambda \sum_{k=1}^{p} \theta_k^2$$

Here, $\hat{\theta_A}$ regularizes the intercept term θ_0, while $\hat{\theta_B}$ does not regularize θ_0. Assume $\lambda = 2$ is fixed for both models. Let MSE_A and MSE_B represent the training MSE for $\hat{\theta_A}$ and $\hat{\theta_B}$, respectively. Which of the following relationships between MSE_A and MSE_B is ALWAYS correct?

A. $MSE_A \geq MSE_B$
B. $MSE_B > MSE_A$
C. $MSE_A = MSE_B$
D. $MSE_A = MSE_B + \lambda$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

What is the effect of pruning a decision tree on its bias, variance, and complexity?

A. Increases bias, decreases variance, decreases complexity
B. Decreases bias, increases variance, increases complexity

C. Increases bias, decreases variance, depends on the splitting rule for complexity
D. Decreases bias, decreases variance, increases complexity

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

We are trying to train a decision tree for a classification task where 0 is the negative class and 1 is the positive class. We are given 8 data points each in pairs of (x_1, x_2) features.

x_1	x_2	y
3	4	1
2	1	0
1	3	1
5	9	0
9	6	1
7	2	1
4	7	0
8	8	1

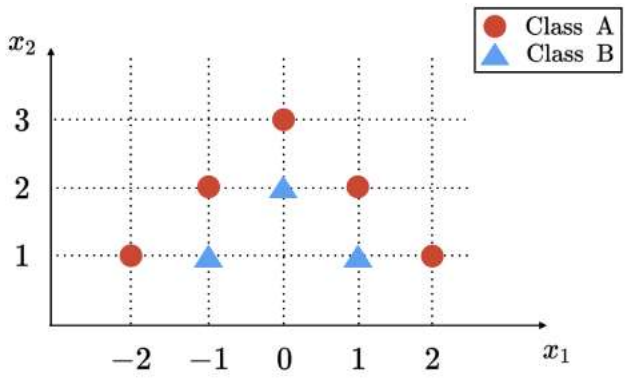
Suppose we decide to split the root node with the rule $x_i \geq \beta$ where $\mathrm{i} = 1$ or 2 . Which of the following minimizes the weighted entropy of the two resulting child nodes.

A. $x_1 \geq 6$
B. $x_1 \geq 3.5$
C. $x_2 \geq 5$
D. $x_2 \geq 3.5$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Recall that in a binary classification task, we want our data to become linearly separable so that we can maximize the performance of our classifier. In many cases, however, our data are not directly linearly separable. As a result, we want to apply some transformation to our data so they will become linearly separable afterwards.
For the following dataset, select all transformations that can make the data linearly separable.



A. $(x_1, x_2) \rightarrow (x_1^2, x_2)$
B. $(x_1, x_2) \rightarrow (x_1, x_2^2)$
C. $(x_1, x_2) \rightarrow (x_1^2, x_2^2)$
D. $(x_1, x_2) \rightarrow (x_2^2, x_1^2)$

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Copyright & Stuff